Robust Certificates in Learning Theory

Combinatorial and Algorithmic Aspects

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Background & Research Questions

Outline

1. Background & Research Questions

- Motivation
- Paper: Proofs as Explanations *[BHPS25]*
- Our Research Questions

2. Examples

3. Certificate Extraction (Algorithmic) *[RQ1]*

~~4. Upper Bound: Tolerance Helly (Combinatorial)~~ *[RQ2]*

5. Summary & Open Questions

The Trustworthiness Problem in ML

Machine Learning models are deployed in high-stakes scenarios. For example, a doctor might use a classifier to decide if a patient is sick.

In such cases, there is a need for developing **trustworthiness**. The doctor would like to know:

Correctness. Is the model correct — under our assumptions (model), and evidence (training data)?

Explainability. Why is the patient classified as sick?

The Model: Proofs as Explanations *[BHPS25]*

Setting. Domain $\mathcal{X}$, labels $\mathcal{Y} = \{0, 1\}$, hypothesis class $\mathcal{H} \subset \mathcal{X}^{\mathcal{Y}}$. Training set $S \subseteq \mathcal{X} \times \mathcal{Y}$.

Definition (Certificate). S is a **certificate** for $z^\star = (x^\star, y^\star)$ if and only if:

1. $\forall h \in \mathcal{H}$ s.t. $h(x^\star) \neq y^\star$ ("bad functions"):

$$\sum_{(x,y)\in S} 1[h(x) \neq y] \;\geq\; 1.$$

2. S is realizable.

$$\exists h \in H : \sum_{(x,y)\in S} 1[h(x) \neq y] \;=\; 0.$$

S serves as a **"proof"** that $h(x^\star) = y^\star$.

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Definition (b-Robust Certificate). S is a b-**robust certificate** for $(x^\star, y^\star)$ if and only if:

1. $\forall h \in \mathcal{H}$ s.t. $h(x^\star) \neq y^\star$ ("bad functions"):

$$\sum_{(x,y)\in S} 1[h(x) \neq y] \;\geq\; b+1.$$

2. S is b-robustly realizable:

$$\exists\, h \in \mathcal{H} \text{ s.t. } h(x^\star) = y^\star : \quad \sum_{(x,y)\in S} 1[h(x) \neq y] \;\leq\; b.$$

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Shortness / Minimality. We call S a **short** b-certificate for $z^\star$ (or *minimal-inclusive*) if no proper subset $C \subsetneq S$ is a b-certificate for $z^\star$.

Certificate in Action: Linear Classifiers

Setting: $\mathcal{H}$ = halfspaces through the origin in $\mathbb{R}^2$,
$b = 0$.
Training set $S = \{p_1, p_2, p_3, p_4\}$.

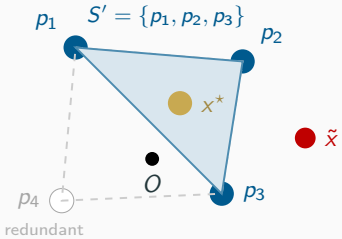
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From **Carathéodory's Theorem** - largest short ceritificate is of size $d + 1$.



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Combinatorial Dimension

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A generalization of the *hollow star number [Bousquet, Hanneke, Moran, Zhivotovskiy — "Proper Learning, Helly Number, and an Optimal SVM Bound", 2020]*
that lower bounds the sample complexity of proper learners.

Q2: Sample Complexity of Certification

Sample Complexity of Pointwise Certification

Data drawn i.i.d. from distribution $\mathcal{D}$ over $\mathcal{X}$ (labeled by $h^\star \in \mathcal{H}$); adversary corrupts $\leq b$ samples. Fixed test point $x^\star$. *How many samples suffice to certify $x^\star$?*

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Result *[BHPS25]*:

$$\Theta\left(\frac{d \log \frac{1}{\varepsilon_{x^\star}} + \log \frac{1}{\delta}}{\varepsilon_{x^\star}}\right)$$ samples suffice *and* are necessary.

($d = \mathrm{VCdim}(\mathcal{H})$, δ = failure probability. Tight upper *and* lower bounds.)

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Analogy: replacing $\varepsilon_{x^\star} \to \varepsilon$ (uniform) recovers the classical PAC sample complexity for the realizable setting.

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Question *(analyzed for $b = 0$)*

Given a sample S that certifies $z^\star$, can we efficiently extract a *minimum-size* sub-certificate $C \subseteq S$ for $z^\star$?

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Our answers:

- The problem is **structurally equivalent** to minimum transversal (aka vertex cover or hitting set) in a hypergraph.
- We give an Oracle-based algorithm which (approximately) solves this task efficiently under finite VCdim or finite s_0.

(Our) Research Question 2 — Combinatorial

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Known lower bound *[BHPS25]:*

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Our answer (upper bound):

$$s_b(\mathcal{H}) \;\leq\; \eta\big(s_0(\mathcal{H}),\, b{+}1\big) \;=\; \begin{cases} O(b^{s_0-1}) & \text{fixed } s_0 \\ O(s_0^{b+1}) & \text{fixed } b \end{cases}$$

via the Tolerance Helly framework of Montejano–Oliveros + Tuza's extremal estimate.

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Examples: s_0 and s_b

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   $s_b(\mathcal{H}) \approx \Theta(d^{b+1})$

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2. **Singletons (Finite Domain):** $\mathcal{H} = \left\{ h_a := 1[x = a] \;\middle|\; a \in \{1, \ldots, N\}, \right\}$

   $s_0(\mathcal{H}) = N.$

   $s_b(\mathcal{H}) = (b+1)(s_0 - 1) + 1 = \Theta(b \cdot s_0)$

Certificate Extraction (Algorithmic)

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3. **Certificate Extraction (Algorithmic)** **[RQ1]**
   - Equivalence: Certificates = Transversals
   - Oracle-Based Algorithms & Guarantees
4. ~~*Upper Bound: Tolerance Helly (Combinatorial)*~~ *[RQ2]*
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Certificate Extraction (CE) Framework

Notation $z_i = (x_i, y_i)$

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Setting. We are given hypothesis class $\mathcal{H}$, a finite labeled sequence $S = (z_1, z_2, \ldots) \subseteq \mathcal{X} \times \mathcal{Y}$ and a point $x^\star$ such that S certifies $z^\star = (x^\star, y^\star)$.

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Assumption. We are given S and $\mathcal{H}|_S$ (the restriction of $\mathcal{H}$ to S) explicitly, e.g. in matrix form:

$$A_{ij}(\mathcal{H}, S) = h_i(x_j).$$

Input Size $= O(|\mathcal{H}|_S| * |S| + |S|)$

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Definition (Hitting Set). $C \subseteq S$ is a **hitting set** of $\mathcal{M}$ if

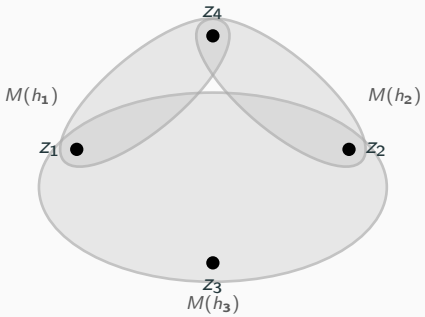
$$C \cap M \;\neq\; \varnothing \qquad \text{for all } M \in \mathcal{M}.$$

Key fact: C certifies $(x^\star, y^\star) \iff C$ is a hitting set of $\mathcal{M}$.

The Mistake Hypergraph

Sample $S = \{z_1, z_2, z_3, z_4\}$, three bad-label hypotheses:

	z_1	z_2	z_3	z_4
h_1	1	0	0	1
h_2	0	1	0	1
h_3	1	1	1	0



Minimal transversals: $\{z_1, z_4\}$, $\{z_1, z_2\}$, $\{z_2, z_4\}$, $\{z_3, z_4\}$

Hypergraphs & The Mistake Hypergraph

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(Mistake Hypergraph = Mistake Collection).

Certificate Extraction $\equiv$ Hypergraph Transversal

Theorem 1 (Structural equivalence).

(Forward) Let $(\mathcal{H}, S, z^\star)$ be a CE instance. Then $C \subseteq S$ is a certificate for $z^\star \iff C$ is a transversal of $\mathcal{M}$: $C \cap M(h, S) \neq \varnothing$ for all $h \in V^-(z^\star)$.

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Consequence. $\mathrm{OPT} = \min\{|C| : C \text{ certifies } (x^\star, y^\star)\} = \tau(\mathcal{M})$, the *minimum transversal number* of $\mathcal{M}$.

Proof: CE $\rightarrow$ Transversal (Theorem 2)

Theorem 1 (Theorem 2 — Forward direction)

$C \subseteq S$ is a certificate for $z^\star$

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*C is a **transversal** of $\mathcal{M}$, i.e., $C \cap M(h) \neq \varnothing$ for all $h \in V^-$.*

Proof sketch.

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- $h \in V^-$ is consistent with C iff $M(h, S) \cap C = \varnothing$ (no mistakes on C)
- So C is a certificate iff every $M(h)$ is hit by C. $\square$

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For every finite hypergraph $\mathcal{M} = (S, \mathcal{E})$ (no empty edge), there exists a certificate-extraction instance $(\mathcal{H}, S, z^\star)$ whose mistake hypergraph is exactly $\mathcal{M}$.

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- Then $M(h_E) = \{(x_i, 0) : i \in E\}$ — recovers $\mathcal{F}$ exactly. $\square$

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- **Bounded rank** r**:** $O(r)$**.**
  If all (minimal) M in $\mathcal{M}$ have $|M| \leq r$, $r-$approximation.

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Goal: Runtime $= O(\mathrm{poly}(|S|))$.

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The oracle: weighted bad-label ERM — given weights $(w_z)_{z \in S}$, returns $h^\sharp \in \arg\min_{h \in V^-} \sum_{z \in M(h)} w_z$.

This is a **separation oracle** for the LP.
By the GLS theorem (ellipsoid method): **LP solved in $O(\mathrm{poly}(|S|))$ oracle calls.**

Oracle-Based Algorithm: Guarantees

Approximation ratio (randomized rounding):

Running $O(\log(1/\delta))$ independent trials and keeping the smallest transversal gives, with probability $\geq 1 - \delta$, a certificate of size

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This ratio is tight.

Lovász (1975): the integrality gap of the standard set-cover LP is $\Theta(\log M)$ for a universe of size M. No LP rounding can beat this in the worst case.

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Bounded VC dimension ($d = \mathrm{VCdim}(\mathcal{H})$):

Sauer–Shelah: $|\mathcal{M}| \leq O(|S|^d)$, so $\log |\mathcal{M}| = O(d \log |S|)$.

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Bounded hollow star number (s_0):

Brute-force over all subsets of S of size $\leq s_0 - 1$ (uniform weights $:= 1$)
gives the **exact optimum** in $O(|S|^{s_0})$ oracle calls.

Certificate Extraction: Wrap Up

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We extended it to learning theory via a separation oracle, and showed upper bounds using the combinatorial dimensions s_0, d.

Open Questions — New Directions

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2. Can the algorithmic process of acquiring S change the picture?

   $\rightarrow$ S sampled i.i.d. from $\mathcal{D}$: does **not** help (**for** $b = 0$)!

   $\rightarrow$ **Active Learning**: learner queries points $x \in \mathcal{X}$ sequentially, adversary answers with labels

   (or via a "Selective Sampling" framework)

Outline

1. Background & Research Questions

2. Examples

3. Certificate Extraction (Algorithmic) [RQ1]

~~*4. Upper Bound: Tolerance Helly (Combinatorial)*~~ *[RQ2]*

- The Question Again
- *b*-Robust Hollow Star Number
- Tolerance Helly Theorem
- Application to Hollow Stars
- Extremal Combinatorics: Critical Hypergraphs

5. Summary & Open Questions

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A word on tightness. *We do not prove a matching lower bound — We conjecture it can be matched using Tuza's extremal construction of critical hypergraphs (a dedicated slide to come).*

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Connection to certificates:

($\Rightarrow$) **Hollow star $\to$ Short Certificate.** Let S be a hollow star set and choose arbitrary $(x, y) \in S$. Then $S \setminus \{(x, y)\}$ is a short 0-certificate for $(x, 1-y)$.

($\Leftarrow$) **Short Certificate $\to$ Hollow star.** Let C be a certificate for $(x^\star, y^\star)$. Then $C \cup \{(x^\star, 1-y^\star)\}$ is a hollow star set.

Definition 3.3 (Robust Realizability with Weights). A weighted sequence $\{(x_i, y_i, w_i)\}_{i=1}^n$ is b-**robustly realizable** by $\mathcal{H}$ if

$$\exists\, h \in \mathcal{H} : \quad \sum_{i=1}^{n} w_i \cdot 1[h(x_i) \neq y_i] \;\leq\; b.$$

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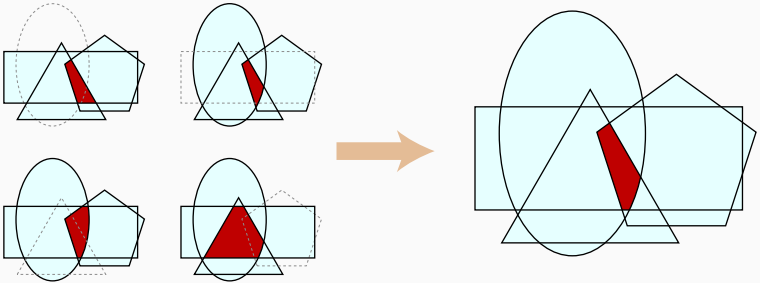
Theorem. $s_b(\mathcal{H}) - 1$ is the worst-case minimal b-robust certificate size. $b = 0$ recovers s_0.

Helly's Theorem

A finite family of convex sets in $\mathbb{R}^d$ has a common point $\iff$ every subfamily of size $\leq d+1$ has a common point. The threshold $\mu = d+1$ is the *Helly number*.

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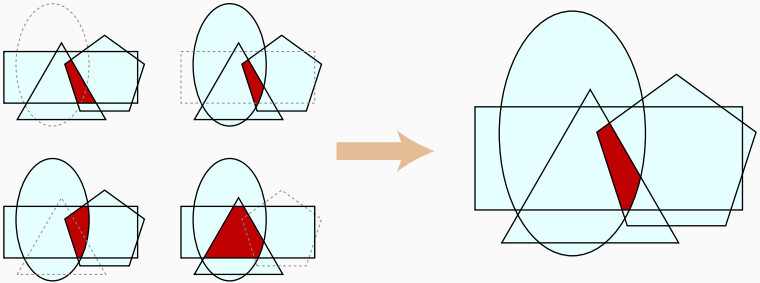


In $\mathbb{R}^2$ ($d{=}2$): every *three* of the sets intersect $\Rightarrow$ all of them intersect.

Image: David Eppstein, Wikimedia Commons (public domain).

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In $\mathbb{R}^2$ ($d{=}2$): every *three* of the sets intersect $\Rightarrow$ all of them intersect.

Image: David Eppstein, Wikimedia Commons (public domain).

Equivalently: the largest minimal-inclusive non-intersecting family has size $\mu = d{+}1$.

The Helly Number of a Property

Generalisation *[Montejano–Oliveros].* Let P be a property of *finite* families that is monotone ($P(F) \Rightarrow P(G)$ for $G \subseteq F$). Its *Helly number* μ is the smallest integer such that: if every subfamily of size $\leq \mu$ has P, then the whole family has P.

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Contrapositive reading (the one we use):

$$\mu \;=\; \text{size of the largest } \textit{minimal-inclusive} \text{ family on which } P \textit{ fails}.$$

Call S *critical* for P iff:

(1) S fails P, and

(2) it is minimal with respect to (1).

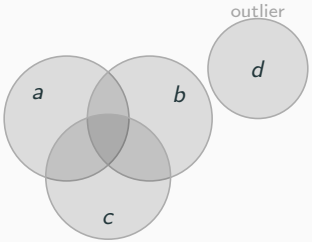
So any critical S has size $\leq \mu$.

Adding Tolerance

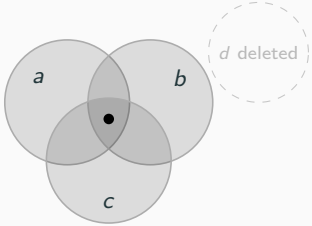
Property with tolerance k *[Montejano–Oliveros].* A family F has property P with tolerance k if there is some $G \subseteq F$ with $|G| = k$ such that $F \setminus G$ has P — i.e. P holds after deleting *some* k elements.

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no common point of a, b, c, d



common point with tolerance 1

The Tolerant Helly Number

Helly number of the b-*tolerant* version of P $=$ largest minimal-inclusive family that is not b-tolerant for P.

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Theorem (Tolerance Helly *[Montejano–Oliveros]*). *If P is monotone with Helly number μ, then the b-tolerant version of P has Helly number at most $\eta(\mu,\ b{+}1)$.*

Here $\eta(\mu, k)$ is the *Erdős–Gallai bound*, unpacked shortly.

Let S be critical for P_b, then $|S| \leq \eta(\mu,\ b{+}1)$.

s_0 is the Helly Number of Realizability

The property: $P(S) \iff$ "S is realizable by $\mathcal{H}$", i.e. $\bigcap_i V(x_i, y_i) \neq \varnothing$.

It is **Monotone:** a subsequence of a realizable sequence is realizable.

Helly number of realizability $=$ **largest size of** S **critical for realizability** $= s_0(\mathcal{H})$**.**

What about a b-robust hollow star set? Is it critical for P_b*?*

b-Robust Hollow Star is Not Directly Critical

Let S be a b-robust hollow star set with $|S| = m + 1$: m *light points* and 1 *star point* $z^\star$.

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Not necessarily. $S \setminus \{z^\star\}$ might be realizable, so S could already have P_b (i.e. S itself is b-tolerantly realizable).

Solution — the uplift. Duplicate $z^\star$ exactly $b{+}1$ times to get $S^\uparrow$.
Now removing the star cluster costs $b{+}1$ deletions $> b$, so $S^\uparrow$ **fails** P_b.

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Issue 2: is $S^\uparrow$ minimal-inclusive for failing P_b?

Not necessarily. If S already failed P_b, then $S \subsetneq S^\uparrow$ also fails P_b, so $S^\uparrow$ is not minimal.

Solution — extract a critical subfamily. **Claim:** there exists $F^\star \subseteq S^\uparrow$ that *is* critical for P_b and contains **all** m light points (and at least 1 star point).

Therefore: $m \leq |F^\star| - 1 \leq \eta(s_0, b{+}1)$.

The Upper Bound

Theorem (main bound). *For every hypothesis class $\mathcal{H}$ and every integer $b \geq 0$,*

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Theorem (main bound). *For every hypothesis class $\mathcal{H}$ and every integer $b \geq 0$,*

$$s_b(\mathcal{H}) \;\leq\; \eta\big(s_0(\mathcal{H}),\, b{+}1\big).$$

Proof in one breath.

1. Realizability is monotone with Helly number s_0
2. The lift $T^{\uparrow}$ has a minimal-inclusive non-b-tolerant subfamily $F^{\star}$ containing all $m{-}1$ light points
3. Tolerance Helly bounds that minimal family: $|F^{\star}| \leq \eta(s_0, b{+}1)$
4. Therefore $|S| = m + 1 \leq |F^{\star}| \leq \eta(s_0, b{+}1)$. □

It remains to say *what η actually is*.

Outline

1. Background & Research Questions

2. Examples

3. Certificate Extraction (Algorithmic) [RQ1]

~~*4. Upper Bound: Tolerance Helly (Combinatorial)*~~ *[RQ2]*

- Tolerance Helly Theorem
- Application to Hollow Stars

↪ **Extremal Combinatorics: Critical Hypergraphs**

5. Summary & Open Questions

Critical Hypergraphs

Transversal number. For a hypergraph G, the *transversal number* $\beta(G)$ is the cardinality of a minimum vertex cover set.

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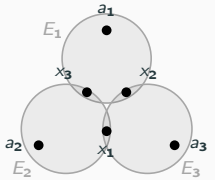
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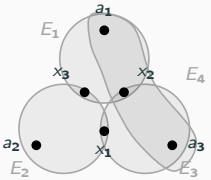
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2-critical



$\beta = 2$; removing any E_i drops it to 1.

not critical



$\beta = 2$ still; removing E_4 leaves it at 2.

The Erdős–Gallai Bound η

$\eta(\lambda, k)$ = **the maximum number of vertices of a k-critical λ-uniform hypergraph.**

Posed by Erdős–Gallai (1961); sharp estimates by Tuza (1989).

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Theorem 2.1 (Erdős–Gallai; Helly-type for transversals). *For a λ-uniform hypergraph G and integer k,*

$$\beta(G) \le k \iff \beta(H) \le k \quad \text{for every sub-hypergraph } H \text{ with } |V(H)| \le \eta(\lambda,\, k{+}1).$$

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A Helly-type theorem in its own — with Helly number $\eta(\lambda, k{+}1)$

The Montejano–Oliveros Graph

The hypergraph $G_\mu(F)$ *[Montejano–Oliveros].* Given a family F with Helly number μ, build a μ-*uniform* hypergraph:

- **vertices** = the elements of F;

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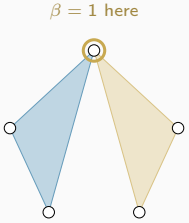
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Deleting b elements so that P holds = a vertex cover of size b. Hence

$$F \text{ **is** } P \text{ **with tolerance** } b \iff \beta(G_\mu) \le b.$$

Criticality: If G_μ is not critical $\to \exists$ a μ-subhypergraph $G'_\mu \subsetneq G_\mu$ with the same $\beta \to G_\mu$ is not minimal.



μ-uniform edges = failing subsets = non realizable sets

Criticality = Minimal-Inclusive

(Excluding the heavy/star point.) Let S be a labeled set. Define the **transversal number**

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$$G(S) \text{ is } (b{+}1)\text{-critical} \iff S \text{ is minimal non-}b\text{-realizable.}$$

Asymptotics

Tuza's sharp estimate *[Tuza 1989]:*

$$s_b(\mathcal{H}) \;\leq\; \eta(s_0, b{+}1) \;<\; \binom{s_0+b}{s_0-1} + \binom{s_0+b-1}{s_0-1}.$$

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Open: show the lower bound explicitly — exhibit classes whose s_b matches $\eta(s_0, b{+}1)$ (e.g. via Tuza's extremal construction).

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We used machinery from **extremal combinatorics** to upper-bound the combinatorial worst-case growth of $s_b(s_0, b)$.

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Remark: Distribution breaks extremality. If S is a minimal b-robust certificate acquired by sampling i.i.d. from a distribution $\mathcal{D}$, then w.h.p. $|S| = \Theta(b \cdot s_0)$, not extremal. *[BHPS25, Theorem 4.1]*.

Summary & Open Questions

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We extended the trustworthiness model of *[BHPS25]* with two main research questions. In both cases the problems are naturally viewed as problems on **hypergraphs** (equivalent to set systems), and both results are **worst-case** and **(almost) distribution-free**.

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Main open direction:

A modified *active-learning* flavor framework for acquiring pointwise certificates — can sequential querying provably reduce certificate size or sample complexity?